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MILITARY SPECIFICATION

NIGHT VISION SIGHT, INDIVIDUAL SERVED WEAPON AN/PVS-4

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Sight, Individual Served Weapon AN/PVS-4.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

H-B-1654	- Brushes, Dusting, Lens and Photographic Negatives
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals
NNN-P-40	- Paper, Lens
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-S-760	- Strapping, Nonmetallic (and Connectors)
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-18	- Batteries, Dry
MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items
MIL-I-49040	- Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-850111	- Housing, Battery
DL-SM-B-850301	- Night Vision Sight, Individual Served Weapon AN/PVS-4
DL-SM-B-850121	- Eyeguard Assembly
DL-SM-B-850316	- Daylight Cover
DL-SM-B-850381	- Adapter, Low Temperature
SM-B-850201	- Case, Carrying and Storage
SM-D-850202	- Bag, Carrying

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Night Vision Sight, Individual Served Weapon AN/PVS4, herein referred to as the sight, is intended for use at night in aimed firing of individual served weapons.

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3.2 First article. When specified in the contract (6.2), the contractor shall furnish first article sights in accordance with 4.4.

3.3 Construction. The sight shall be constructed in accordance with DLSM-B-850301 and as specified herein.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor, shall be subject to all provisions of this specification, and shall conform to MIL-P-11268.

3.5 Major components. The following components shall be constructed in accordance with applicable drawings, specifications, and as specified herein.

3.5.1 Objective lens assembly. The objective lens assembly shall conform to MIL-L-49366.

3.5.2 Eyepiece lens assembly. The eyepiece lens assembly shall conform to MIL-L-49368.

3.5.3 Eyeguard assembly. The eyeguard assembly (referred to herein as the eyeguard) shall conform to DLSM-B-850121.

3.5.4 Low temperature adapter. The low temperature adapter shall conform to DLSM-B-850381.

3.5.5 Image intensifier assembly. The image intensifier assembly shall conform to MIL-I-49040.

3.5.6 Battery housing. The battery housing shall conform to DLSM-B-850111 and as specified herein.

3.5.7 Daylight cover. The daylight cover shall conform to DLSM-B-850316.

3.5.8 Battery. The battery shall be a BA 1567/U and shall conform to MIL-B-18.

3.5.9 Brush. The brush shall conform to H-B-1654.

3.5.10 Lens paper. The lens paper shall conform to NNN-P-40.

3.5.11 Case. The case, carrying and storage, shall conform to SM-B-850201.

3.5.12 Bag. The bag, carrying, shall conform to SM-D-850202.

3.6 Operation and environment.

3.6.1 Eyeguard operation. With the eyeguard vertical and at -48°C, the eyeguard shall open (depress) when a 2000, ±10 gram weight is placed axially

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on the eye flange and shall close (expand) when the weight is reduced to 100, ±10 grams. At +23°C, the eyeguard shall open (depress) when a 1600, ±10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100, ±10 grams. At +23°C and with the eyeguard open (depressed) the minimum open position of the 5 leaves shall be 1.875 inches.

3.6.2 Low temperature adapter. The low temperature adapter shall not be damaged by storage in any temperature from +23°C to -62°C and shall operate as specified herein at any temperature from +23°C to -54°C.

3.6.3 Resolution. The minimum sight resolution shall be within one element number of the resolution of the image intensifier being used in the sight at the time of testing while using a USAF 1951 Resolving Power Test Target.

3.6.4 Field of view. The sight shall have a minimum field of view of 253 milliradians.

3.6.5 Vibration. The sight shall withstand simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion) with the frequency being varied between 10 and 55 hertz (Hz).

3.6.6 Weapon shock. The sight, mounted and zeroed (see 6.3) on the M16 Rifle, shall not shift in zero more than ±0.50 milliradian or be damaged (see 6.3) when subjected to the vibration during 500 rounds of rifle firing.

3.6.7 Weight. The weight of the sight with batteries but without daylight cover shall not exceed 4.0 pounds.

3.6.8 *Controls and adjustments*. All controls and adjustments shall operate smoothly without binding throughout their complete range of movement at all operating temperatures specified herein.

3.6.8.1 Eyepiece diopter focus. With the diopter adjustment control set at 0 ±1/2 diopter, the image of the projected reticle shall be in sharp focus.

3.6.8.2 Range focus. The sight shall provide a minimum focus adjustment range from a near focus of 25 meters to infinity. The dynamic torque to turn the range focus control shall be between 5 and 15 inch-pounds.

3.6.8.3 Tube brightness control. The tube brightness control shall vary the gain of the image intensifier tube from OFF to maximum gain, and control the power to the reticle brightness control.

3.6.8.4 Reticle brightness control. Brightness of the reticle pattern shall be continuously variable over the range from OFF to full brightness. The pattern shall be visible and the numbers legible when viewing a solid white target having a brightness of 3.3×10^{-3} footlamberts minimum.

3.6.8.5 Reticle movement. Two external control knobs shall control the position of the reticle pattern viewed by the observer using the sight. One control shall move the reticle pattern in azimuth; another control shall move

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the reticle pattern in elevation. An adjustment made by turning one control shall not affect the position established by the other control. Adjustment will be apparent to the operator by tactile indexing with audible clicks. Each click of the control knob shall displace the reticle pattern 0.25, ±0.04 milliradian. Total movement of the reticle pattern shall be a minimum of 2.5° in any direction from the optical axis. The torque required to adjust the reticle position shall be between 0.5 and 4.0 inch-pounds. The reticle pattern shall remain in focus throughout the full range of movement of the range focus control. The reticle pattern shall not exhibit rotation greater than 1° from its proper orientation (see figure 1).

3.6.9 High temperature storage and operation. The sight shall not be damaged (see 6.3) by storage in any temperature from +23°C to +68°C and shall operate as specified herein at any temperature from +23°C to +52°C.

3.6.10 Low temperature storage and operation. The sight and low temperature adapter shall not be damaged by storage in any temperature from +23°C to -62°C and shall operate as specified herein at any temperature from +23°C to -54°C.

3.6.11 Altitude. The sight shall not be damaged when operated at a pressure equivalent to 10,000 feet altitude above sea level.

3.6.12 Temperature shock. The sight shall not be damaged after being subjected to temperature changes between +68°C and -62°C in a 5 minute period.

3.6.13 Humidity. The sight shall not be damaged when operated in an atmosphere with a relative humidity of 95 percent, ±5 percent over a temperature range of +27°C to +45°C and shall operate after storage up to +65°C minimum with 95 percent, ±5 percent relative humidity.

3.6.14 Immersion. The sight shall not be damaged after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.6.15 High intensity shock. The sight shall not be damaged by a sequence of shocks applied in each direction along each of 3 mutually perpendicular axes (horizontal, vertical, and optical axes when the reticle pattern is erect). The shocks shall be half sine pulses and shall have a time duration of 4 milliseconds, ±5 percent. Shock pulses shall have a peak amplitude of 100 g's, ±15 g's (see 6.3).

3.7 Reliability. The sight shall have a minimum Mean Time to Failure (MTTF) of 1000 hours when operated under the reliability test conditions specified herein.

3.8 Government loaned property. Unless otherwise specified in the contract or purchase order, the Government will provide the following items for use in the quality assurance tests of Section 4 (see 6.2):

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NSN	Nomenclature	Quantity
1005-00-073-9421	Rifle M16	3
1305-00-926-3930	Cartridge 5.56 mm Ball A071	As required (see 6.2)
5855-00-138-2386	25 mm Microchannel Inverter Tube in conformance with MIL-I-49040	10 each

3.9 Marking. Unless otherwise specified, markings on the sight and all components shall conform to MIL-M-13231.

3.10 Workmanship. Unless otherwise specified, workmanship of the sight and ancillary equipment shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Components and materials inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (4.4).
b. Quality conformance inspection (4.5).
c. Packaging inspection (4.9).

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order, inspections shall be performed as specified herein. Tolerances on specified illumination levels shall be plus or minus 10 percent. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer. When a collimator is used, equivalent focal length (EFL) of the collimator shall be at least six times the EFL of the objective lens under test. The clear aperture of the collimator shall be 6 inches or greater. Tolerance on 2854 K color temperature shall be ±200 K. References within environmental requirements and tests to "initial measurements", "operate", "pretest", or "performance check during test", shall be defined as the resolution test 4.6.3. External power of 2.6, ±0.1 volts dc (Vdc) shall be used during environmental testing.

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4.4 First article inspection.

4.4.1 Inspection. Each sight shall be inspected in process in accordance with table I. Presence of one or more defects shall be cause for rejection of that sight.

TABLE I. Inspections.

Defects	Requirement paragraph	In process
101. Performance requirements specified in notes appearing on drawings.	3.3	X
102. Linear or diametral tolerances of 0.002 inch or less.	3.3	X
103. Angular dimension having a tolerance of 15 minutes of arc or less.	3.3	X
104. Machine finishes of 32 micro-inches RMS or finer.	3.3	X
105. Tolerances of position or form 0.002 inch or less.	3.3	X
106. Screw thread Class 3 or better.	3.3	X
107. Load requirements for springs.	3.3	X
108. Lubrication requirements including dry lubricants finishes.	3.3	X
109. Material hardness when specified on drawings.	3.3	X

4.4.2 Tests. Following successful completion of the inspections specified in 4.4.1, each first article sight shall be subjected to all inspections in table II. Failure of any inspection shall be cause for rejection of that sight. Inspections may be conducted in any order except resolution shall be performed last.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Eyeguard operation	3.6.1	4.6.1
Low temperature adapter	3.6.2	4.6.2
Sight assembly		
Field of view	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Weapon Shock	3.6.6	4.6.6
Weight	3.6.7	4.6.7
Eyepiece diopter focus	3.6.8.1	4.6.8.1
Range focus	3.6.8.2	4.6.8.2
Tube brightness control	3.6.8.3	4.6.8.3
Reticle brightness control	3.6.8.4	4.6.8.4
Reticle movement	3.6.8.5	4.6.8.5
High temperature storage and operation	3.6.9	4.6.9
Low temperature storage and operation	3.6.10	4.6.10
Altitude	3.6.11	4.6.11
Temperature shock	3.6.12	4.6.12
Humidity	3.6.13	4.6.13
Immersion	3.6.14	4.6.14
High intensity shock	3.6.15	4.6.15
Resolution	3.6.3	4.6.3
Reliability	3.7	4.7

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Sampling shall be in accordance with MIL-STD-105, Inspection Level II, combined AQL one percent defective. Following completion of the inspections of table I, each sight that has passed the inspections shall be inspected per table III. Failure of any sight to pass all the tests of table III shall be cause for rejection of that sight.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3.3, 3.4	4.4.1
Tube brightness control	3.6.8.3	4.6.8.3
Reticle brightness control	3.6.8.4	4.6.8.4
Reticle movement	3.6.8.5	4.6.8.5

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4.5.2 Group B inspection. Sights which have passed the inspections specified in 4.5.1 shall be formed into lots (see 6.3.5). Sample sights shall be selected and subjected to the inspection of table IV. Sampling shall be in accordance with MIL-STD-105, combined AQL of 1.5 percent for the group, inspection level II. Inspections shall be performed in the order listed.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Eyepiece diopter focus	3.6.8.1	4.6.8.1
Resolution	3.6.3	4.6.3
Range focus	3.6.8.2	4.6.8.2

4.5.3 Group C inspection. Group C inspections shall be conducted on sights selected from lots which have passed the inspections in 4.5.2. The sample(s) shall be subjected to the inspections specified in table V. Samples shall be selected in accordance with 4.5.3.1. Tests may be performed in any order except that in Subgroup 1, immersion shall follow vibration and weapon shock, and in Subgroup 2, humidity shall be last.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
High temperature storage and operation	3.6.9	4.6.9
Low temperature storage and operation	3.6.10	4.6.10
Vibration	3.6.5	4.6.5
Weapon shock	3.6.6	4.6.6
Immersion	3.6.14	4.6.14
Subgroup 2		
High intensity shock	3.6.15	4.6.15
Temperature shock	3.6.12	4.6.12
Humidity	3.6.13	4.6.13

4.5.3.1 Sampling for Group C inspection.

Subgroup 1. For this subgroup, 3 sights shall be selected at random and shall be subjected to the group C, Subgroup 1 tests specified in table V. The sample shall be selected from the total quantity in lots passing group B tests each 30 days or from each 100 sights in lots passing group B tests, whichever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot.

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Subgroup 2. For this subgroup, 1 sight shall be randomly selected from the first quality conformance inspection lot to pass group B tests. For subsequent group C, Subgroup 2 inspections, one (1) sight shall randomly be selected every third month from the inspection lots passing group B tests that month.

4.5.4 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order (see 6.2).

4.5.5 Disposition of Group C inspection units. Any sights which have been subjected to group C testing shall be subjected (after refurbishment, if necessary) to the test of tables III and IV prior to acceptance. Failure of a sight of any test of tables III and IV shall be cause to reject that sight.

4.5.6 Group D inspection. Group D test shall be conducted on sights selected from units which have passed the inspections of 4.5.1. Tests shall be conducted in accordance with table VI.

4.5.6.1 Sampling for Group D inspections. The acceptance test shall be performed on the first quality conformance inspection lot and every other month or 100 sights thereafter, whichever occurs first. Sample size shall be 10.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.7	4.7

4.5.6.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or order (see 6.2).

4.5.6.3 Reinspection of Group D sample units. Unless otherwise specified, sample units which have been subjected to group D inspection may be accepted on contract or order provided all damage is repaired and the units are re-subjected to and pass the tests of tables III and IV.

4.6 Test methods.

4.6.1 Eyeguard operation.

4.6.1.1 Low temperature compression/expansion. Place the eyeguard in an environmental chamber at -48°C, +3°C for 2 hours. Without removing the eyeguard from the chamber, place a 2000, ±10 gram weight axially on the eye flange. Reduce the weight to 100, ±10 grams. Failure of the eyeguard to open (depress) within 5 seconds after adding the 2000 gram weight and to close after reducing the weight to 100 grams shall constitute failure of this test.

4.6.1.2 Room temperature compression/expansion. At +23°C, repeat the inspection of 4.6.1.1 using a 1600, ±10 gram weight. Failure of the eyeguard

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to open (depress) within 3 seconds after adding the 1600 gram weight and to close after reducing the weight to 100, ±10 grams shall constitute failure of this test.

4.6.1.3 Leaf separation. During inspection of 4.6.1.1 and 4.6.1.2, measure the leaf separation. Failure of the leaf separation to be 1.875 inches or greater shall constitute failure of this test.

4.6.2 Low temperature adapter. The low temperature adapter shall be exposed to the low temperature storage and operation inspection in 4.6.10 while attached to a sight, using either a hard power source or a BA 1567/U battery. Failure of the sight to produce the characteristic green glow of the intensifier thru the eyepiece, or failure of the low temperature adapter to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Resolution. Place a 100 percent contrast USAF 1951 Resolving Power Test Target (black on a clear background) in the focal plane of a collimator having a known focal length. Back illuminate the target with a 2854 K white light of such intensity that the brightness of the target through the collimator is 3.3×10^{-3} footlamberts. Place the sight with the range focus set for infinity looking into the collimator. Determine the smallest pattern which can be resolved. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Field of view. Place two dark vertical lines on a wall or on a pattern in a collimator with 253 milliradian separation with respect to the sight. With the sight stationary, observe through the eyepiece. Inability to see both lines at the edge of the field of view at one time shall constitute failure of this test.

4.6.5 Vibration. The sight shall be subjected to Method 201A of MIL-STD-202. The test after vibration shall be resolution. Operation during vibration shall not be required. The induced vibratory force shall be applied in each of 3 mutually perpendicular planes one of which is perpendicular to the optical axis. The duration of vibration shall be five minutes in each plane. Damage to the sight or failure to pass the resolution test shall constitute failure of this test.

4.6.6 Weapon shock. This test may be conducted during daylight hours with a daylight cover. The sight shall be mounted on an M16 rifle by means of a weapon adapter bracket (see 6.2) and shall be sighted on a target. The sight shall be zeroed (see 6.3) and all hardware and connectors shall be checked to insure that no parts are loose. No adjustments, tightening of parts or repairs of any kind, shall be permitted after the initial zeroing is completed and test firing cycles have begun.

4.6.6.1 Quality conformance testing. The weapon shall be fired in 4 to 10 round bursts, until 100 rounds have been fired. During the firing, evidence of loss of power to image intensifier shall constitute failure of this test. After 100 rounds have been fired, the sight shall be inspected for damage and the position of the reticle in the sight with the relative positioning prior to firing the 100 rounds. Any evidence of damage or loss of zero exceeding 0.5 milliradian shall constitute failure of this test.

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4.6.6.2 First article testing. The firing and examination of 4.6.6.1 shall be repeated sequentially until 500 rounds have been fired. Failure to conform to the acceptance criteria of 4.6.6.1, or evidence of loss of zero exceeding 0.5 milliradian when compared to the relative position prior to firing the 500 rounds, shall constitute failure of this test.

4.6.7 Weight. Weigh the sight on a scale which can be read to within ±0.1 pound with an accuracy within ±1.0 percent. Failure to meet the requirement of 3.6.7 shall constitute failure of this test.

4.6.8 Controls and adjustments.

4.6.8.1 Eyepiece diopter focus. With the system operating, observe the reticle pattern through the eyepiece with a diopter scope of 3 power or greater set at 0 diopter. Failure to obtain sharp focus of the reticle pattern within +1/2 to -1/2 diopter as indicated in the eyepiece scale shall constitute failure of this test.

4.6.8.2 Range focus.

a. For the infinity focus requirement, place a point source of light in the focal plane of a collimator. Observe the point source with the sight. Adjust the range focus in and out, expanding the image size on both sides of best focus. Inability to increase the size of the image on both sides of both sides of best focus shall constitute failure of the test.
b. For the minimum focus requirement, place a point source of light at a distance of 25.0, +0.5 meters in front of the objective. Adjust the range focus and observe that the point source can be brought into sharp focus. Inability to bring the 25 meter target into sharp focus shall constitute failure of this test.
c. Measure the dynamic torque required to rotate the range focus. Failure to comply with the requirements of 3.6.8.2 shall constitute failure of this test.

4.6.8.3 Tube brightness control. Attach the sight to a test bench so that a collimator target can be projected directly into the objective assembly. Back illuminate the target located at the focal plane of the collimator at 1×10^{-5} footcandles. Turn the Tube Brightness control clockwise past the detent position. Open the eyeguard, look into the eyepiece, and adjust the Tube Brightness for maximum brightness. Measure the dynamic torque required to rotate the tube brightness control. Failure of the tube brightness to vary as the control is turned, or to meet the requirements of 3.6.8.3 shall constitute failure of this test.

4.6.8.4 Reticle brightness control. With the sight, observe a solid white target of brightness 3.3×10^{-3} footlamberts in a collimator. Measure the dynamic torque required to rotate the reticle brightness control. Failure to clearly distinguish the reticle pattern, or to meet the requirements of 3.6.8.4 shall constitute failure of this test.

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4.6.8.5 Reticle movement. (See 6.2) With the same test setup as 4.6.8.3 and with a test target per figure 2, turn on the reticle brightness control and adjust for optimum illumination of the reticle pattern. Adjust the range focus and diopter ring for a sharp image of the reticle pattern.

4.6.8.5.1 Adjustment accuracy. With the reticle center on the crosshairs of the test target, turn the reticle adjustment 20 clicks to the right and observe the position of the reticle center relative to the two lines A and B. Repeat measurement, moving the reticle 20 clicks to the left of the target center. Repeat these measurements using the up and down reticle adjustment. Failure of the center of the reticle pattern to fall between the two lines A and B for any of the four observations shall constitute failure of this test.

4.6.8.5.2 Adjustment range. With the reticle pattern at its center of movement, aline the test target center crosshairs with the reticle. Turn the azimuth adjustment until the center of the reticle reaches one extreme of the 5° circle and then back to the opposite side of the circle. Return the reticle to the center of the circle. Repeat this procedure using the elevation adjustment. Failure of the reticle to traverse the 5° circle or evidence of one adjustment affecting the other or evidence of binding or slipping of the reticle shall constitute failure of this test.

4.6.8.5.3 Torque. Using a torque wrench and adapter, rotate the elevation adjustment, first clockwise and then counterclockwise, and observe the readings. Repeat this procedure for the azimuth adjustment. Failure of any of the four torque readings to fall between 0.5 and 4 inch-pounds or failure to hear the audible click or feel the tactile indexing shall constitute failure of this test.

4.6.8.5.4 Alinement. Place a target (figure 2) containing vertical and horizontal reference lines vertically in the focal plane of the collimator. Verify that the objective lens is level with respect to the mounting pad. Center the reticle pattern on the target crosshairs. Failure of the reticle pattern to maintain alinement with the reference lines within one degree shall constitute failure of this test.

4.6.8.5.5 Clarity. Focus the viewer at 25 meters distance and illuminate the scene at approximately 1×10^{-5} footcandles. Refocus the viewer on a target at infinity using the collimator. Failure of the reticle pattern to be sharp and clear or all numbers legible at both 25 meters and infinity focus shall constitute failure of this test.

4.6.9 High temperature storage and operation. The sight shall be subjected to Test Method 501, Procedure I, MIL-STD-810. Operating temperature (Step 4) shall be $+52^{o}C$ with the sight turned on for 3 hours. Storage temperature (Step 2) shall be $+68^{o}C$ for 6 hours duration. The "operate", of Step 5 shall be defined as the ability to observe a green glow from the eyepiece (with the eyeguard present but detached) from the exterior of the chamber thru a window. Failure to operate per Steps 5 and 7 or pass inspection of Step 8 shall constitute failure of this test.

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4.6.10 Low temperature storage and operation. The sight shall be subjected to Test Method 502, Procedure I, MIL-STD-810. Storage temperature (Step 2) shall be -62°C for 9 hours and operating temperature (Step 4) shall be -54°C for 9 hours. The "operate" of Step 5 shall be defined as the ability to turn all controls within their respective torque requirements and to observe a green glow from the eyepiece (with the eyeguard present but detached) from the exterior of the chamber thru a window. Failure to operate per Steps 5 and 7 or pass inspection of Step 8 shall constitute failure of this test.

4.6.11 Altitude. The sight shall be subjected to Test Method 500, Procedure I, MIL-STD-810 except that the minimum chamber pressure shall be equivalent to 10,000 feet above sea level. Operate the sight (Step 2) for 3 hours. Damage (see 6.3) or failure to operate per Step 2 shall constitute failure of this test.

4.6.12 Temperature shock. The sight shall be subjected to Test Method 503, Procedure I, MIL-STD-810. The maximum temperature (Steps 1 and 4) shall be +68°C and the minimum temperature (Step 2) shall be -62°C. Any evidence of damage or failure to operate or to pass the inspection of Step 9 shall constitute failure of this test.

4.6.13 Humidity. The sight shall be subjected to Test Method 507, Procedure II, MIL-STD-810, except that only two continuous 48 hour cycles shall be required. During the last 5 hours of each cycle (Step 6), operate the sight and check for a green glow in the eyepiece. Failure to observe the green glow in Step 6, operate per Step 8, or pass the inspection per Step 9 shall constitute failure of this test.

4.6.14 Immersion. The sight shall be subject to Test Method 512, Procedure I, MIL-STD-810, and any evidence of damage (see 6.3) shall constitute failure of this test.

4.6.15 High intensity shock. The sight shall be subjected to Test Method 516, Procedure IV, MIL-STD-810 in accordance with figure 516.2-2. The shock shall be half sine pulses having a peak amplitude of 100, ±15 g's and a time duration of 4 milliseconds, ±5 percent. Damage (see 6.3) or failure to pass the operational test shall constitute failure of this test.

4.7 Reliability. The contractor shall conduct the reliability test on 10 sights. The test shall be performed under the test conditions specified herein. The test shall terminate when each sight either accumulates 1500 hours of operation ("ON" time) or experiences an image intensifier tube failure (non-repairable failure).

4.7.1 Test conditions. The reliability test shall be conducted in accordance with Test Level C of MIL-STD-781. During operation, each sight shall have a target brightness of not less than 1×10^{-4} footlamberts with the gain control set at maximum. The target brightness shall be increased to 1×10^{-2} footlamberts for 5 seconds once each hour of operation. A light source of 10 to 20 footcandles shall be within the field of view for 3 seconds once during each hour of operation. The increased target brightness and

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light source within the field of view shall not occur within 1 minute of one another. Sufficient instrumentation shall be provided to ensure immediate recognition of a catastrophic failure (including continuous recording of input current to each sight) as well as a change in the relative output brightness of each sight. Performance shall be measured, at a minimum, once every 112 hours of operating time by at least the following tests or inspections:

a. System resolution
b. Damage, system and tube assembly
c. Luminance gain of the tube assembly
d. Equivalent background input of the assembly
e. Signal-to-noise of the tube assembly

A performance test failure shall be presumed to have occurred immediately after the last successful measurement or inspection. A sight shall be removed from the test for failure analysis upon the recognition of a failure. If the failure is in the image intensifier tube (non-repairable), the sight shall not be returned to test. All other failures shall be repaired and the sight returned to the test. Each sight and image intensifier tube assembly shall be submitted to Group A inspection and the inspection of 4.5.1 if and when the sight has accumulated 1500 hours of operation. The image intensifier assembly shall be submitted to Group A of MIL-I-49040. The following specific requirements of MIL-STD-781 shall apply:

1. Design and performance tests
2. Detail reliability test procedure
3. Inspection
4. Test facilities
5. Selection of equipment
6. Installation of equipment
7. Testing the setup
8. Determination of compliance
9. Failure actions
10. Failure categories
11. Analysis of failure
12. Failure confirmation
13. Verifying repair
14. Corrective action
15. Restoration of failed equipments
16. Final report

4.7.2 Failure definition. A failure is defined as the inability of the sight and/or tube assembly to meet the requirements of Group A inspection or the inspection of 4.5.1. This includes the inability of the tube assembly to meet the Group A requirements of specification MIL-I-49040.

4.7.3 Accept/reject criteria. The accept/reject determination shall be made upon completion of the test as specified herein. Acceptance criteria shall be that the average of the total operating times to either the initial failure or 1500 hours, whichever occurs first, of each sight is 1000 hours or more. Rejection criteria shall be that the mean time to failure is less than 1000 hours.

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4.8 Government loaned property. GLP will be furnished as specified in the contract (see 6.2).

4.9 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the Group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Night Vision Sight, Individual Served Weapon, AN/PVS-4 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Night Vision Sight, Individual Served Weapon, AN/PVS-4 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Night vision sight, individual served weapon; low temperature adapter; M60 machine gun bracket; M60 reticle cell; bag, carrying; brush; and paper, lens. Each item shall be individually unit packed method III as follows: Fold bag, carrying to smallest dimension. Place brush and paper lens into a bag suitable to protect each item. Position all components into recesses provided in case, storage. Close cover and secure latches. Place the case within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of Night Vision Sight, Individual Served Weapon, AN/PVS-4, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more, and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.2 Level B. A quantity of Night Vision Sight, Individual Served Weapon AN/PVS-4, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Night Vision Sight, Individual Served Weapon AN/PVS-4, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The AN/PVS-4 is intended for use at night for aimed firing of individual served weapons.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Which of the following major components are to be purchased and the quantities involved:

Batteries, BA 1567/U per MIL-B-18
Low temperature adapter per DL-SM-B-850381
M60 Mounting bracket per DL-SM-B-850341
M67 Mounting bracket per DL-SM-B-850351
M72 Mounting bracket per DL-SM-B-850361
M79 Mounting bracket per DK-SM-B-850371
M16 Mounting knob per DL-SM-B-850501
Bag, Cloth, Laminated Vinyl-Nylon SM-D-850482-1
Bag, Cloth, Nylon (for Arctic Use) SM-D-850482-2
Reticle M60, M14 per drawing SM-D-850490-2
Reticle M67 per drawing SM-D-850490-4
Reticle M72A1 per drawing SM-D-850490-5
Reticle M16, M16/XM203, M79 per drawing SM-D-850490-7

c. If the image intensifier tube per MIL-I-49040 is to be purchased:

(1) To obtain the sight with Image Intensifier Tube, specify the use of drawing SM-D-850100-1
(2) To obtain the sight without Image Intensifier Tube, specify the use of drawing SM-D-850100-2

d. When the GLP (see 3.8) will be delivered by the Government and the number of cartridges to be supplied.
e. That the contractor will supply sufficient batteries MIL-B-18, BA 1567/U for all inspections. The batteries used for inspection are not to be delivered.
f. Necessary actions by the contractor in the event of a Group C or D failure (see 4.5.4 and 4.5.6.2).
g. Prior to packing and packaging, each sight shall be examined for absence of batteries.
h. Desired level of preservation, packaging and packing (see 5.1, 5.2).
i. Quantity and schedule for first article testing for all line items (see 4.4).

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j. The contractor is responsible to furnish sufficient M16 mounting knobs and daylight covers for the weapon shock test (see 4.6.6).
k. Associate lots for Quality Control Inspection with the monthly requirements established in the contract schedule.
l. When the image intensifier is obtained as part of the sight, the reliability test of MIL-I-49040 shall be waived and the reliability test of this specification performed. When the sight is obtained less image intensifier or if the intensifier is Government Loaned Property, the reliability test of this specification shall be waived.
m. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
o. This requirement and test may be waived if test is performed on the lens unit as a separate item.

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; leakage or condensation of moisture within the sight; degradation in input or output characteristics; and any failure of the control knobs to perform their function.

6.3.2 Zero. A known and specified relationship between the sight aiming point and the strike of the bullet.

6.3.3 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

6.3.4 Room temperature. Room temperature is defined as +23°C, ±3°C.

6.3.5 Lot size. A lot is defined as those sights successfully completing Group A inspection for any one month.

Custodian:
Army - ER

User Activity:
Army - AR

Preparing activity:
Army - ER

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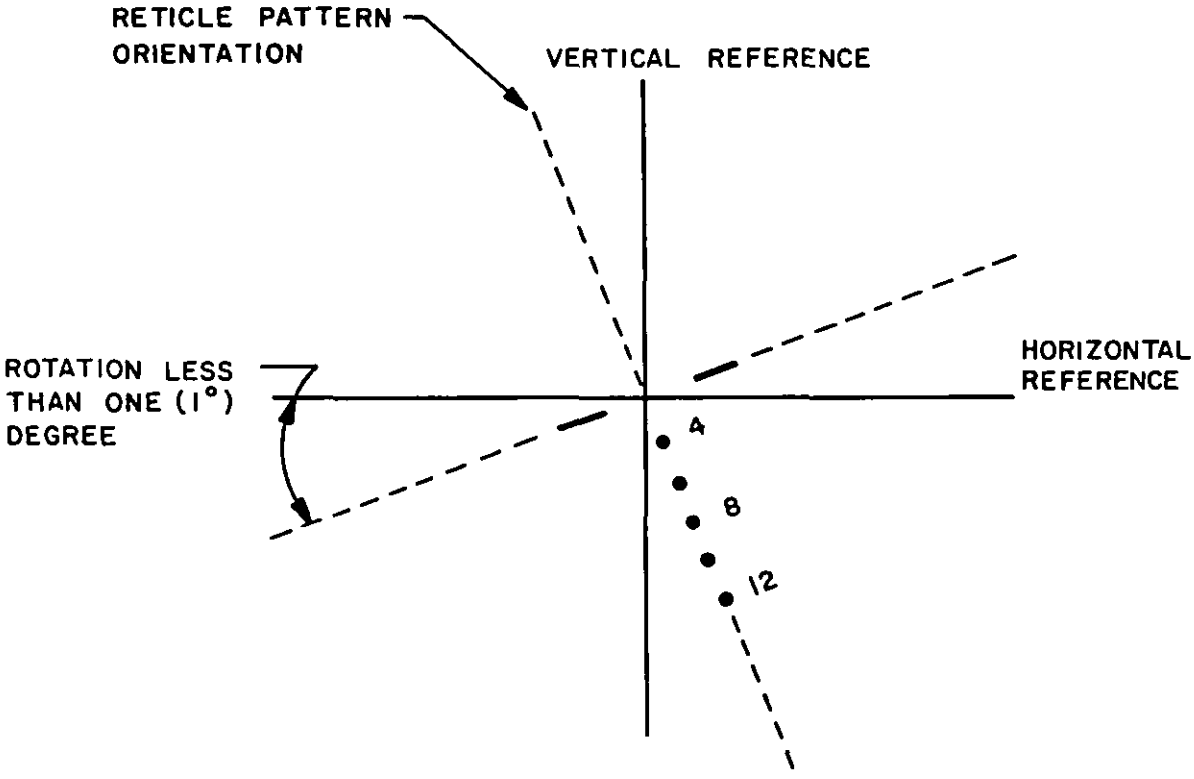
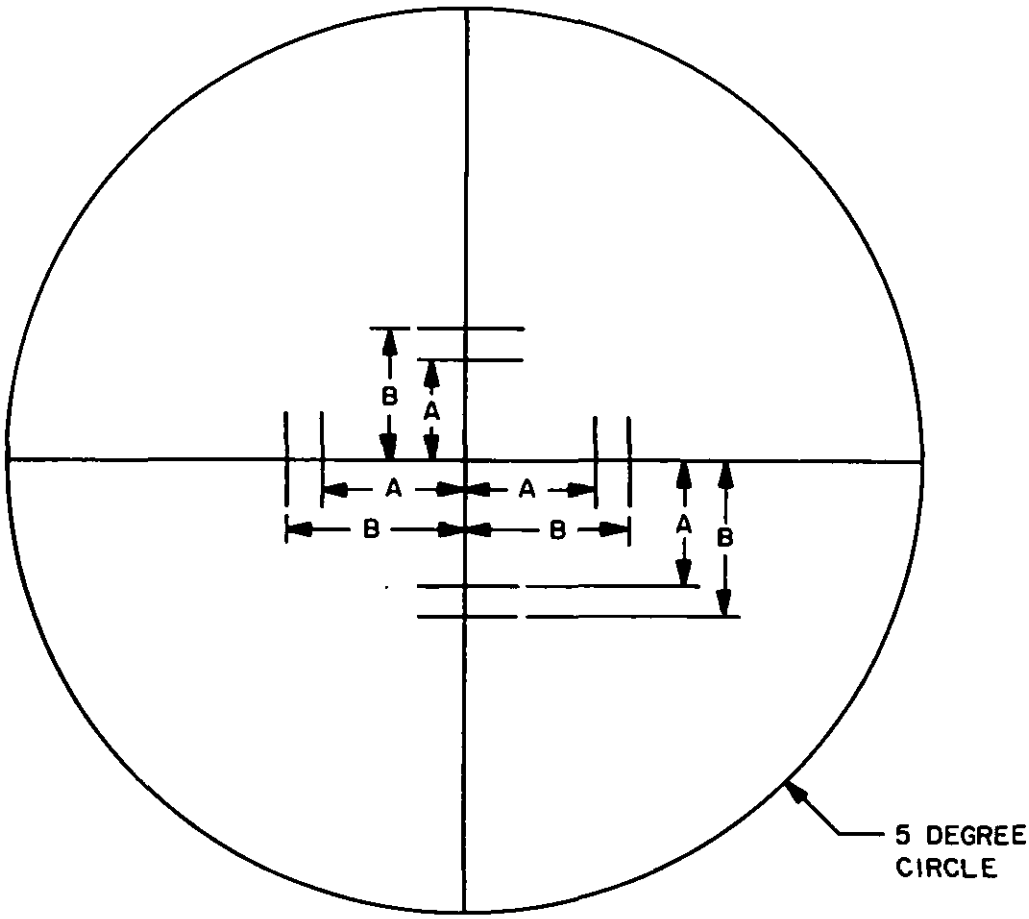


FIG. 1. RETICLE PATTERN ORIENTATION

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NOTE :

1. DIMENSION OF A EQUALS : F X 0.00425
2. DIMENSION OF B EQUALS : F X 0.00575
3. DIMENSION OF CIRCLE EQUALS : .0873 X F
4. F = FOCAL LENGTH OF COLLIMATOR

FIG. 2. TEST TARGET

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